Md. Raiyanul Islam

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RESEARCH INTEREST

Software Supply Chain Security, DevSecOps Automation, Computer Vision, Artificial Intelligence

RESEARCH EXPERIENCE

Academic Mentorship ***Mar 2025 – Aug 2025***

Department of Computer Science and Engineering, Dhaka International University, Dhaka, Bangladesh

Topic: ***A Deep Learning Approach to Predict and Monitor Student Engagement in Academic Environments***

- Led a group of students to design a weak-supervision pipeline to automatically generate a large labeled dataset for real-time classroom engagement detection, leveraging YOLO-based pose detection, Face Mesh, and heuristic rules using EAR, MAR, gaze vectors, and cosine similarity.
- Supervised training of a YOLOv11m engagement classifier predicting five distinct classes directly from raw video frames, achieving 0.73 mAP@0.5

Undergraduate Research ***Jan 2022 – Feb 2023***

Khulna University of Engineering & Technology (KUET), Khulna, Bangladesh

Thesis: ***Phrase Level Bangla Sign Language Recognition using Keypoints from Hand Gesture Video***

- Developed a phrase-level Bangla Sign Language (BSL) recognition system that classifies 27 phrases from hand gesture video using MediaPipe keypoints and an LSTM, reaching 92.07% accuracy on our own dataset.
- Recorded a custom dataset of 1,134 hand gesture videos representing 27 Bangla phrases (single- and two-handed signs) and contributed to implementing the LSTM classifier.
- Designed angle features from MediaPipe hand keypoints in place of raw coordinates; on the Pakistani PkSLMNM dataset (7 phrases) the pipeline reached 90.98% accuracy against 82.66% reported by the dataset authors.

EDUCATION

Bachelor of Science in Computer Science and Engineering | (CGPA 3.64 out of 4) ***Jan 2018 – March 2023***

Khulna University of Engineering & Technology (KUET), Khulna, Bangladesh

- Relevant Coursework: *Machine Learning, Artificial Intelligence, Image Processing*

PUBLICATIONS

1. S. M. M. M. Mahin, **M. R. Islam** and S. M. M. Ahsan, "Phrase Level Bangla Sign Language Recognition using Keypoints from Hand Gesture Video," *2023 International Conference on Next-Generation Computing, IoT and Machine Learning (NCIM)*, Gazipur, Bangladesh, 2023, pp. 1-6 [Recipient of Best Paper Award]

WORK EXPERIENCE

Software Engineer (R&D) | *WellDev, Bangladesh* ***Sept 2025 – Present***

- Built a pipeline for security and non-security dependency updates, with Trivy and Renovate in GitHub Actions, CycloneDX SBOMs in Dependency-Track, and DefectDojo for deduplication, audit history and Slack and Jira alerts.
- Archived Renovate reports in a GCS bucket and connected OpenCVE to DefectDojo to follow CVEs for chosen vendors.
- Replaced the old internal notification bot, which sent every feed update, with alerts only for vulnerability and release updates on libraries a project uses, cutting irrelevant notifications.
- Compared Snyk, Renovate and Trivy on cost, role, CI fit, output formats, per-team reporting and maintenance, and dropped Snyk on cost (free plan: 200 open-source tests a month).
- Built a Claude Code workflow (MCP, Jira, Slack, Google Drive) that checks requests against company licensing and software policies, logs decisions in Jira and Google Sheets, and lets an admin approve or steer.
- Wrote Claude Code skills for onboarding and offboarding, connected to Jira, Google Drive and GitHub, that recommend access changes and apply them through API calls only after an admin with the right role approves in Jira, logging each decision.

Lecturer | *Dept. of Computer Science and Engineering, Dhaka International University, Dhaka* ***Oct 2023 – Aug 2025***

- Delivered lectures, graded student exams, and gave feedback tied to Course and Program Learning Outcomes.
- Supervised and mentored one thesis group and two final-year project groups and served as an academic counselor for two batches, providing comprehensive research guidance, academic support.

Trainee Machine Learning Engineer | *Bista Solutions Ltd., Dhaka, Bangladesh* ***Aug 2023 – Oct 2023***

- Fine-tuned the OpenAI GPT-3.5-turbo model to enhance NLQ-to-SQL conversion efficiency by 14%.

TECHNICAL PROJECTS & OPEN-SOURCE CONTRIBUTIONS

1. **polyscan: Static Code Coupling Analysis** ***Sep 2026***

Tools: Go, Git, static analysis

- Contributed a merged fix to ludo-technologies/polyscan (PR #132, closes #118) to the Go coupling analyzer: type names matching more than one declaration are now left unresolved with a warning instead of being silently dropped or arbitrarily assigned; added tests.

2. **ChatSQL** ***Aug 2023 – Oct 2023***

Tools: Python, Langchain, LLM, OpenAI, PostgreSQL

- Developed an application focusing on personalized database querying through natural language incorporating context history in the conversation to generate and process accurate SQL queries.
- Optimized the performance of NLQ-to-SQL conversion by fine-tuning the OpenAI GPT-3.5-turbo model, addressing real-world database querying challenges for production level databases.

3. **FaceRecog** ***Jan 2021 – Jun 2021***

Tools: Python, sqlite, face-recognition, dlib, PyQt5

- Engineered an end-to-end real-time facial recognition system for automated attendance tracking.
- Utilized a secured authorization pipeline to prevent proxy attendance using eye blink detection approach.
- Implemented automatic and manual attendance marking, and comprehensive data management for faculties and students of an entire institutional organization.

4. **BreadBlogs** ***Dec 2022 – Jan 2023***

Tools: Python, FastAPI, postgres database

- Developed the back-end components of a scalable and blog application using REST architecture.
- Implemented JWT authentication and repository pattern following best practices and clean coding.

TECHNICAL SKILLS

Frameworks	: OpenCV, Tensorflow, Pytorch, Scikitlearn, LangChain, Keras, FastAPI
DevSecOps and Security Tools	: Dependency-Track, Trivy, DefectDojo, CycloneDX SBOM, Renovate, GitHub Actions
Databases & Tools	: PostgreSQL, MySQL, SQLite, Firebase, Docker, Git, Claude Code, MCP
Programming languages	: Python, Java, C, C++, Go, Rust
Documentation	: Microsoft Office, Google Workspace

SCHOLARSHIP & AWARDS

- **Best Paper Award** ***Aug 2023***
  International Conference on Next-Gen Computing, IoT & Machine Learning, Gazipur, Bangladesh
- **Technical Scholarship** ***2022***
  Khulna University of Engineering & Technology (KUET), Khulna, Bangladesh
- **Dean's Award** ***Oct 2022***
  Khulna University of Engineering & Technology (KUET), Khulna, Bangladesh

LANGUAGE PROFICIENCY

IELTS Score**: 7.5** (L: 8.0 | R: 7.5 | W: 6.5 | S: 7.0) ***Aug 2025 – Aug 2027***

LEADERSHIP & CO-CURRICULAR ACTIVITIES

Workshop Instructor ***Jul 2024 – Dec 2024***

Dhaka International University, Dhaka, Bangladesh